

Developing an AI-powered personal endurance sports coach

Airflow Summit 2026
Bas Harenslak

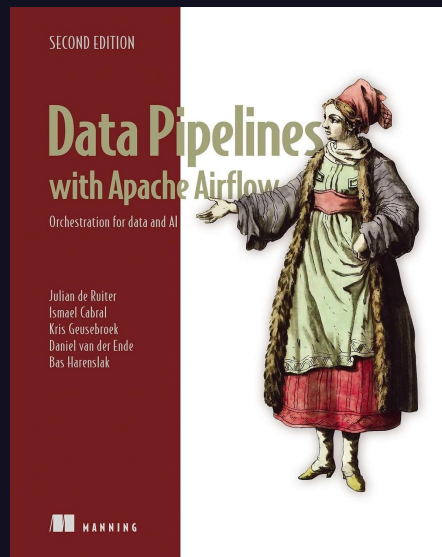
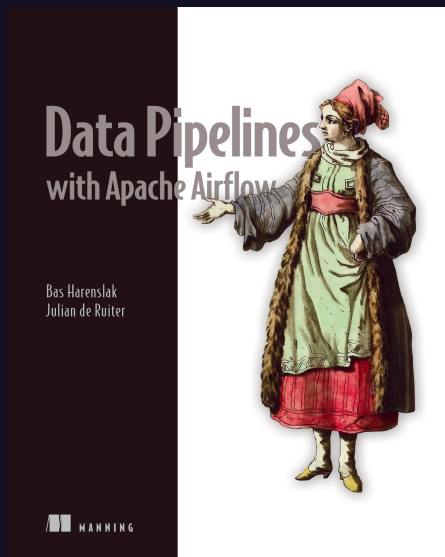
ASTRONOMER



About me

Bas Harenslak

- Staff Architect at Astronomer
- Co-author of book Data Pipelines with Apache Airflow
- Committer on the Apache Airflow project

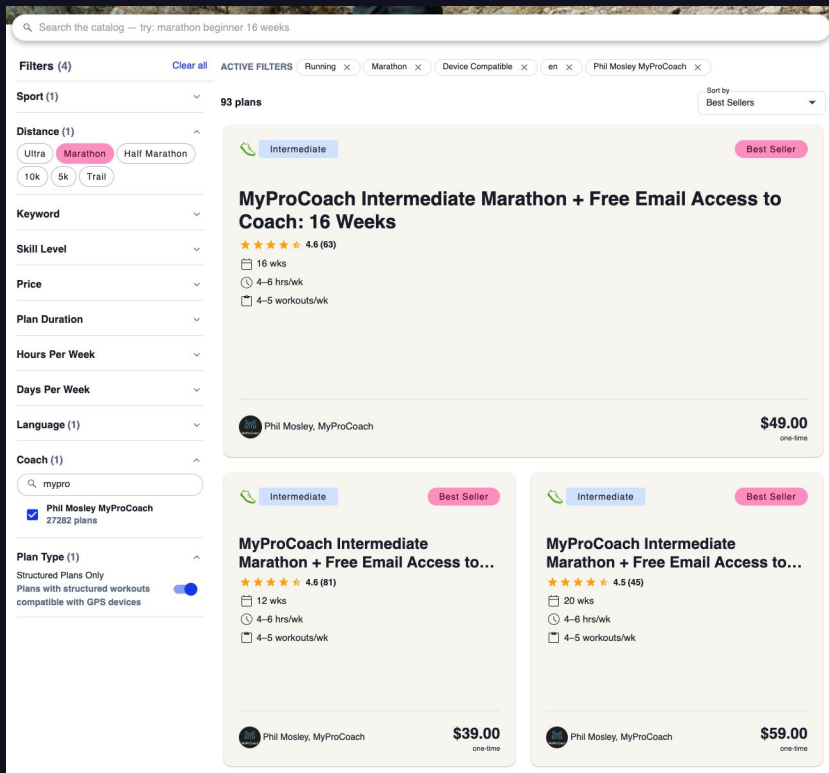


About me



ASTRONJMER

My first training program



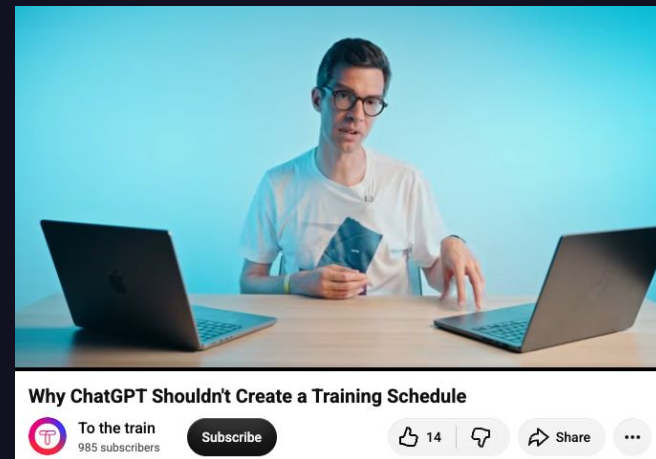
- Planned from start to finish
- No adjustments

From self-coached to personal coach

- Training plan adaptation (fatigue, illness, injuries, travel, etc.)
- Prioritization
- Mental support
- Other parts of endurance sports (nutrition, gear advise, etc.)



AI coach?



MCP



"Write a training plan"
Reactive, not proactive
Inconsistencies



AI coach goals

1. Go beyond the “here’s my data, give me a summary/report”
2. Responsive, not reactive
3. Ability to chat and act on the conversations



First AI steps



Notes in
Google Doc

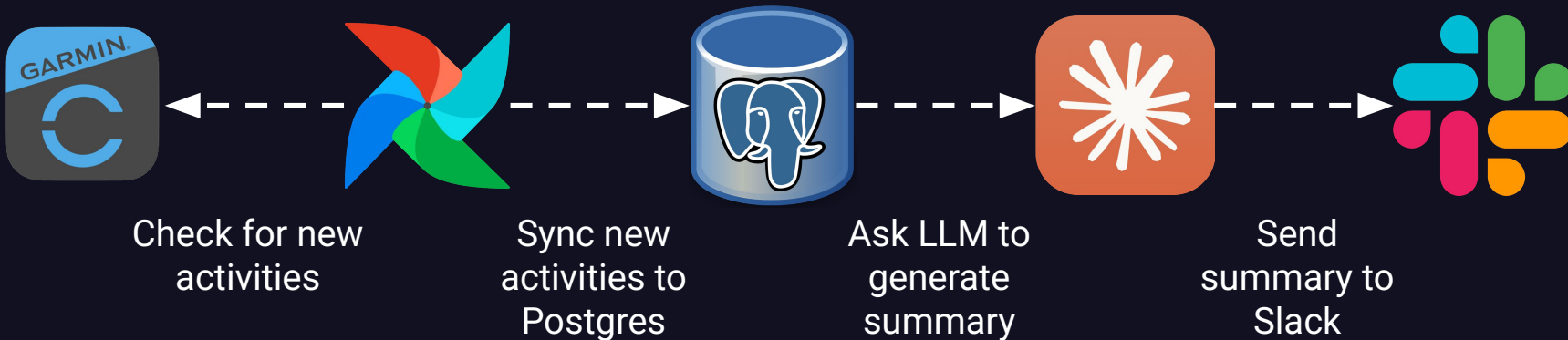


1. Download doc
2. Upload to ChatGPT
3. Ask questions e.g.
 - "What did you think of my training?"
 - "I got stomach cramps, what should I try different next training?"



ChatGPT

First coded approach



AI Trainer APP 11:26 AM

Progressive 10km – Fri 7 Aug. ~49:30 for 11.1 km 🏃

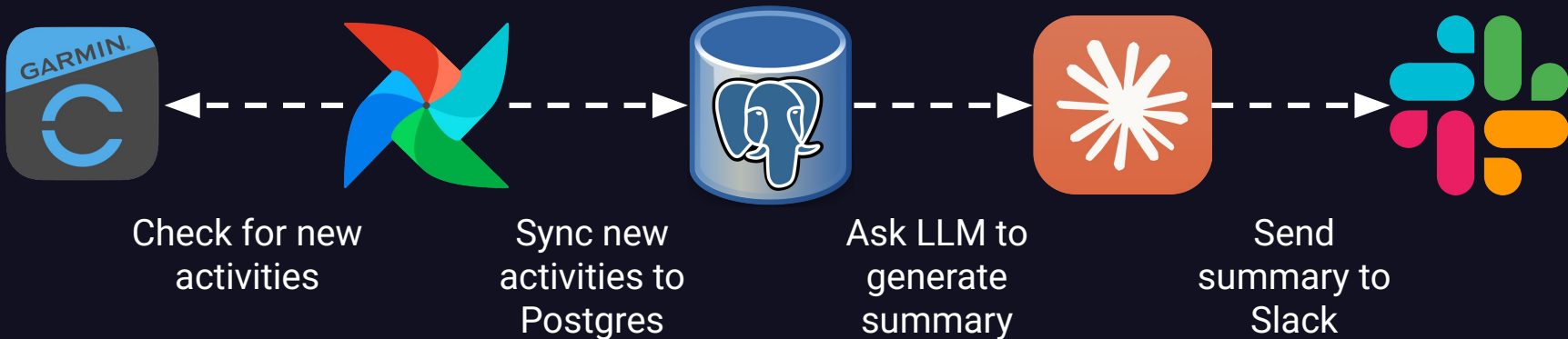
Really nice execution on this one. A progressive run is all about controlled build, and the numbers say you did exactly that.

- Your average pace worked out to about **4:28/km** across 11.1 km at an average HR of just **143**. For a progression that finishes fast, keeping the average that low tells me you started genuinely easy and let the effort come to you rather than forcing it early.
- For context, your recent easy runs sat at 148 HR (30 Jul) and 140 HR (yesterday's Tower Hamlets easy). Landing a progressive at 143 avg – while covering more ground and finishing quicker – is a strong sign of good aerobic control.
- This came the morning after an easy run and off a fairly loaded week (400s, long run, swim, strength, two rides). Handling a progression that cleanly on cumulative fatigue is a good durability marker.

One thing worth flagging: your recovery has been a bit choppy – Thursday night's sleep scored 62 (fragmented, low REM) and sleep need bumped up to 520. HRV is still balanced (78, comfortably in range), so nothing alarming, but keep an eye on getting a solid night in before your next quality session.

Good, disciplined running here – exactly what this session was meant to be.

First coded approach



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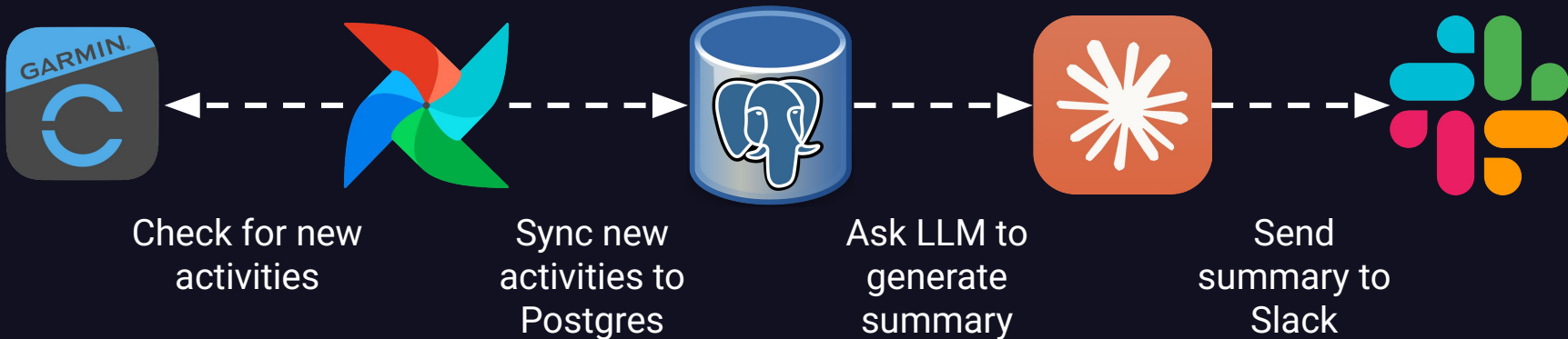
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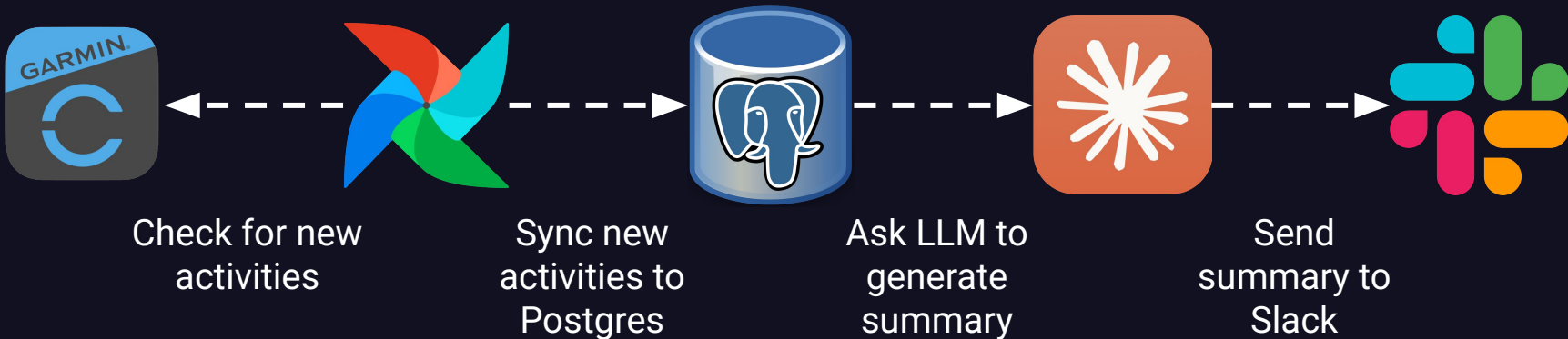
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Summary

First coded approach



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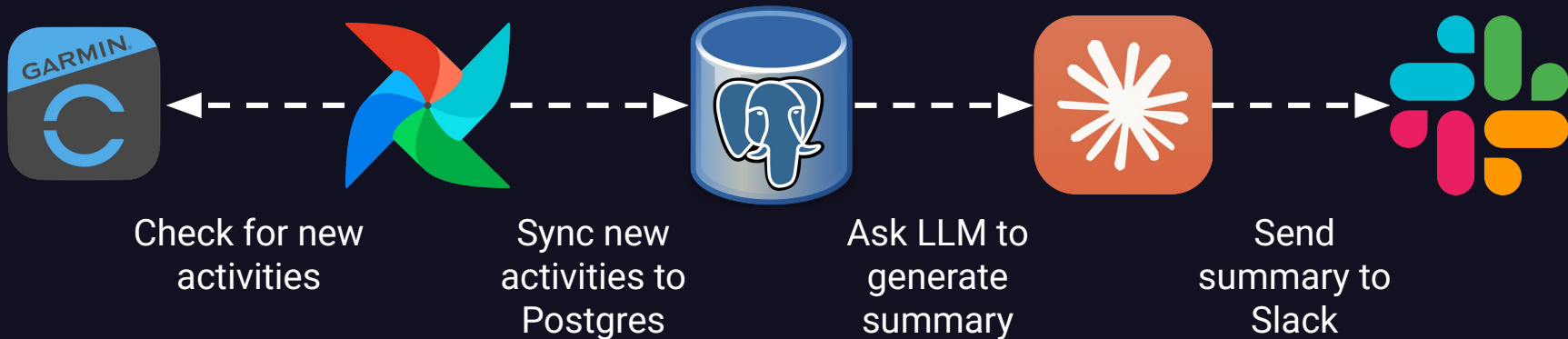
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Highlights

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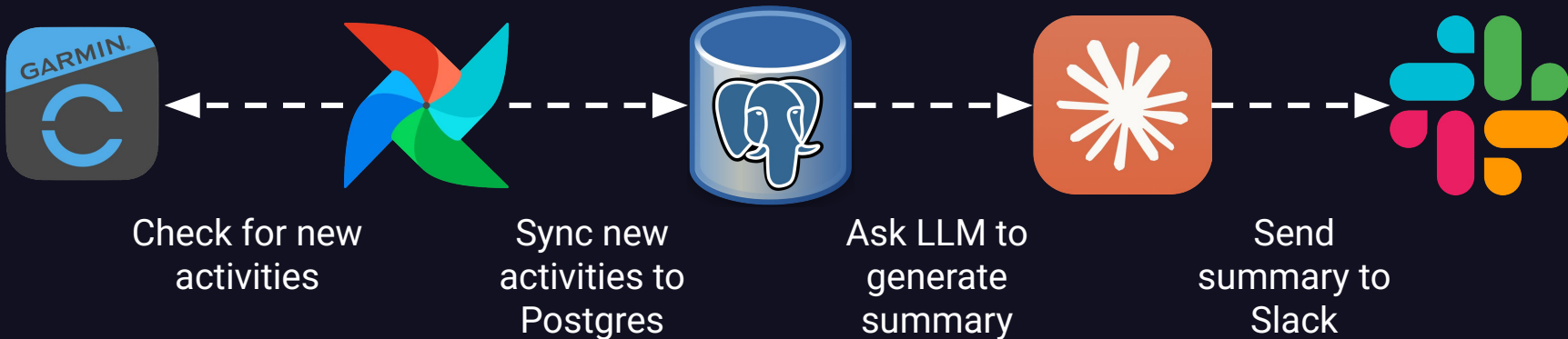
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Attention: focus on good sleep

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Motivation

Asking an LLM for a summary using Airflow

Install `apache-airflow-providers-common-ai`

Install extra toolsets e.g. `apache-airflow-providers-common-ai[anthropic,skills,sql]`

```
from airflow.providers.common.ai.operators.agent import AgentOperator
from airflow.providers.common.ai.toolsets.sql import SQLToolset

compose_message = AgentOperator(
    task_id="compose_message",
    prompt="""
        Here is the latest activity, recent activity history, and recent sleep/HRV:

        {{ task_instance.xcom_pull(task_ids='sync_activities') }}
        {{ task_instance.xcom_pull(task_ids='get_athlete_context') }}
    """,
    llm_conn_id="pydanticai",
    system_prompt=f"""
        You're an endurance sports coach.
        Your athlete just completed a new activity. Provide a small summary.

        Read the new activities and provide your thoughts. For example:
        - Do you think it went well, relative to what the session appears to be for?
        - Anything unique about this activity that stands out worth mentioning?

        {SLACK_FORMATTING_INSTRUCTIONS}
    """,
    toolsets=[
        SQLToolset(db_conn_id="ai_trainer_db")
    ],
)
```

toolsets =
Additional capabilities
and data access

Agent usage is *optional*

Asking an LLM for a summary using Airflow

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`system_prompt =`
General instructions

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        {SLACK_FORMATTING_INSTRUCTIONS}
    """,
    toolsets=[
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    ],
)
```

`prompt =`
Instructions for this run

Initial AI response

 **AI Trainer** APP 10:15 AM
Retrieved 1 new activities from Garmin.

Session Summary — "Leidschendam-Voorburg Wegwielrennen" 🚴

What it was: A ~2h33m, 87 km road race/group ride in flat Dutch terrain (only 232 m of climbing) in warm conditions (20–28°C). Garmin labelled it **Tempo** with a TSS of 140 and Intensity Factor 0.74 — this was a genuinely hard, sustained aerobic effort, most likely a mass-start race given the name and the power profile.

How it went — very well This was a strong, race-appropriate ride:

- **Normalized power 217 W vs. average 208 W** — the modest gap ($VI = 1.04$) shows a well-managed, steady effort rather than a ragged surge-fest. Impressive for a flat race where wind and pack dynamics usually cause spiky power.
- **Big top-end punch when it mattered:** peak 648 W, with 5s at 588 W and 1-min at 294 W. That's classic road-racing signature — sitting in, then covering moves/sprints. 143 s combined in power zones 6–7 confirms repeated hard accelerations.
- **Aerobic Training Effect 4.8 ("Highly Impacting")** with only minor anaerobic load — exactly what you'd expect from a long tempo/race effort.

What stands out:

- **HR stayed remarkably controlled** — avg 131, max only 147, with essentially zero time in HR zone 5 despite those big power spikes. In 28°C heat, that low, stable HR points to solid aerobic fitness and good pacing (VO_{2max} estimate holding at 59). Your cardio is comfortably absorbing the load.
- **~2 L estimated sweat loss** in the heat — worth confirming you matched that with intake; late-race power holding (7200s power 205 W) suggests you fuelled/hydrated well.
- Left/right balance 53/47 — a slight left bias worth a passing glance over time, nothing to act on now.

Pointer for next time: The effort was well-paced and heat-managed, so nothing to fix. If this was a race, the takeaway is that you had aerobic headroom (HR never maxed) — on a flatter/faster course you could afford to spend a bit more in the pack-surge zones. For the next few days, treat this as a big load day (TSS 140) and prioritise easy recovery before your next quality session.

Nice, disciplined ride — this is what good race-day pacing looks like. 🍌

This was a solo ride, not a group ride and not a race.

If the AI trainer had checked my past rides, it should have know that this was an easy ride.

Giving more context: implementing memory

Example messages from athlete to coach:

"Race went well today until I got stomach cramps."

"Felt tired today."

"First upcoming race is 10 mile. What is a realistic target for that?"

Giving more context: implementing memory

Athlete messages

"Race went well today. Everything went well until I got stomach cramps. From that moment I had to switch between walking and running. Here's what I ate/drank:"

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Message

"Race went well today until I got stomach cramps. From that moment I had to switch between walking and running. Here's what I ate/drank:"

Step 1: Extract observations

Observation

During the run, cramps/stitches began just past 5 km in the right lower abdomen immediately after taking the first run gel.

Step 2: Build memories

Memory

Recurring race-day GI issue: right-lower-abdomen cramping that consistently appears on the RUN of triathlons (not on stand-alone rides or shorter events), typically after the first run gel. Walking relieves it instantly; cramp returns on resuming running.

Extracting observations

"Raced ... today.
Everything went well until I got stomach cramps.
From that moment I had to switch between walking and running.
Here's what I ate/drank:"

➔ read_conversation

PythonOperator

✓ success

➔ extract_observations

AgentOperator

✓ success

➔ persist_observations

PythonOperator

✓ success



```
extract_observations = AgentOperator(  
    system_prompt="""  
        Extract coaching observations from incoming athlete events.  
  
        An event represents a single piece of new information, such as:  
  
        * A completed activity  
        * A conversation  
        ...  
  
        An observation should satisfy at least one of the following:  
  
        * Reveals something about the athlete's physical or mental state  
        * Identifies an unusual or noteworthy occurrence  
        ...  
  
        Do not create observations for routine or expected events.  
  
        Examples that SHOULD NOT become observations:  
  
        * Completed a run.  
        * Slept 7 hours.  
        ...  
  
        Examples that SHOULD become observations:  
  
        * Athlete skipped a key workout without explanation.  
        * Repeated calf cramps while running.  
        ...  
    """,  
)
```

Indicate agent what to extract

Extracting observations

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Everything went well until I got stomach cramps.
From that moment I had to switch between walking and running.
Here's what I ate/drank:"

➔ read_conversation

PythonOperator

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➔ extract_observations

AgentOperator

✓ success

➔ persist_observations

PythonOperator

✓ success



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Provide examples of right and wrong

Extracting observations

"Raced ... today.
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Here's what I ate/drank:"

Message from athlete to coach

◆ read_conversation

PythonOperator

✓ success

◆ extract_observations

AgentOperator

✓ success

◆ persist_observations

PythonOperator

✓ success

Extracted observation

During the run, cramps/stitches began just past 5 km in the right lower abdomen immediately after taking the first run gel.



Curating memories from a set of observations

Collection of append-only observations:

During the run, cramps/stitches began just past 5 km in the right lower abdomen immediately after taking the first run gel.

During run, athlete developed a right-side stitch around 10km on an incline, then progressive stomach cramping from half-marathon distance onward, forcing run/walk.

The athlete felt light abdominal pressure after the swim that came and went on the bike, then strong right-lower-abdomen pressure right after starting the run following the first gel.

Recurring GI cramping is right-sided and lower-abdominal (ileocecal region), consistently appears on the run of triathlons.



Memories have a time component:

Some memories are **permanent**: “Prefers outdoor over indoor running.”

Other memories are **temporary**: “No current injuries.”

1. Extract new memories
 - Categorize stability:
 - Acute: hours to days
 - Phase: weeks e.g. small injury recovery
 - Chronic: months e.g. GI issues
 - Permanent: e.g. lactose intolerance
 - Extract validity interval [`to` - `from`]
2. Evaluate existing memories

Curating set of memories from observations

Collection of append-only observations:

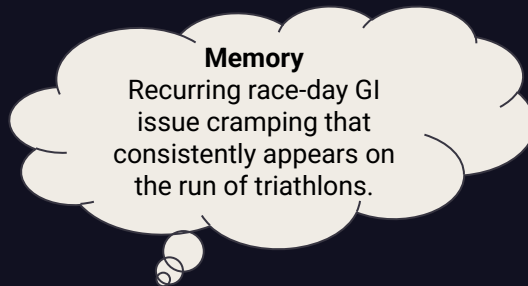


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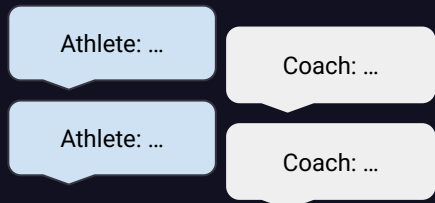


Memory

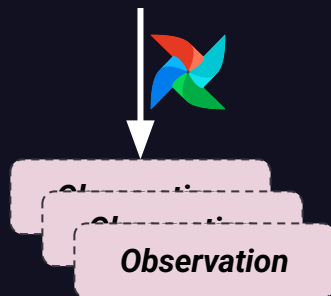
Recurring race-day GI issue cramping that consistently appears on the run of triathlons.

Curating set of memories

Athlete and coach conversation

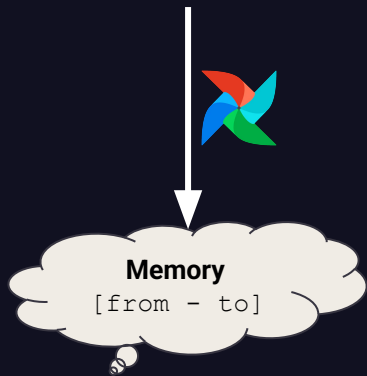


Step 1: Extract observations



Feed back memories into coaching responses

ASTRONOMER Step 2: Build memories



Applying memories in conversations

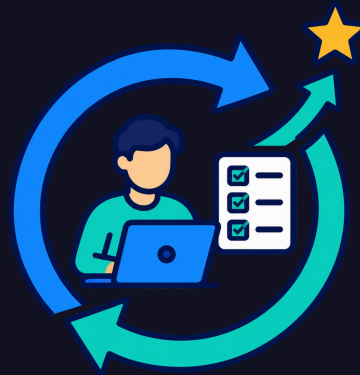
```
AgentOperator(  
    task_id="reply",  
    prompt="""  
        The conversation so far:  
        {{ task_instance.xcom_pull(task_ids='load_conversation_context').history  
    }}  
  
        What you currently know about this athlete (higher priority first):  
        {{ task_instance.xcom_pull(task_ids='load_active_memory') }}  
    """,  
    system_prompt="""  
        You're a triathlon coach.  
  
        Your athlete just messaged you.  
        Answer the message and use available training data when relevant.  
        Keep the reply concise.  
    """,  
)
```

Proactive messaging

Implementing memories greatly improved the message quality. However, responding to activities/athlete messages is **reactive**.

A human coach could reach out **proactively**.

For example, on a lingering injury the coach could check in, ask “how are you feeling?”, and potential change the training plan.



Proactive messaging

```
AgentOperator(  
    system_prompt="""  
        When a memory raises an open question set its `proactive question` to a short question.  
        For example "I see you trained somewhere new. Until when are you travelling?".  
  
        For an injury or pain memory that should trigger recurring follow-up, choose a cadence from the injury described.  
        As a guide:  
  
        - Minor fresh soreness or a small niggle usually merits about 24-72 hours  
        - A likely strain, tendon flare, or pain that may affect training decisions often merits about 3-7 days  
        - A major injury or a slow-moving recovery such as a fracture usually merits about 1-3 weeks  
  
        Leave it null when no recurring injury check-in is needed.  
    """,  
)
```



Any proactive questions due?



If yes: initiate conversation



Costs

Human coach costs 100-400 USD per month.

AI coach costs per month:

- Airflow 30\$
- Postgres DB + other infra 30\$
- Base AI pipeline cost (assuming 2 activities/day): 31.95\$
 - Claude Opus 4.8:
 - 5\$/M input tokens
 - 25\$/M output tokens
- Cost increases with more conversations

Scaling to more messages (total cost per month):

- No communication: 91.95\$
- 1 message per week: 93.67\$
- 10 messages per week: 109.15\$
- 100 messages per week: 263.95\$
- 1000 messages per week: 1811.95\$
- 10000 messages per week (+1 per minute): 17291.95\$

Conclusions & challenges

- Airflow works very well for periodic data gathering and AI analysis. Data is continuously loaded, evaluated, and curated into a set of memories for agent domain knowledge.
- LLMs are stateless: you need to provide all the context with every request to make informed decisions. Balancing the right level of context can be trial-and-error. Calling an LLM is easy, but don't expect great answers when letting your LLM freely search all data. The data plumbing to structure data still requires considerable effort.
- Backfills (generating observations on years of athlete history) led to hitting agent max_tokens limits → required chunking requests and increasing `agent_params={"model_settings": {"max_tokens": ...}}`. Symptom: agent crossed max_tokens and returned empty result. This was technically a valid response in my code, leading to no failure but missing records.
- Runaway LLM: sometimes small changes (e.g. change time boundaries to date without data) would lead to the LLM querying the database until the (default) request_limit of 50 was hit.

The 2026 Apache Airflow® Survey is Live!

- Takes just 7 minutes
- Get a free AI Orchestration Fundamentals Certification Exam or DAG Authoring Certification (normally \$150 each)



Questions

ASTRONOMER

